

(Theoretical) Foundations of Reinforcement Learning

ETH Zürich, Spring 2025

Lecture 5: Policy Gradient Methods II (Theory)

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Outline

Recap: Policy Gradient Methods

Performance Difference Lemma

Global Convergence of Policy Gradient Methods

Exact PG under Direct Parameterization

Global Convergence of Natural Policy Gradient Methods

Tabular NPG with Softmax Parametrization

NPG with Function Approximation

Beyond PG/NPG

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Policy Optimization

Policy Optimization (Episodic Reward)

$$\max_{\theta} J(\pi_{\theta}) := \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \mid s_0 \sim \mu, \pi_{\theta} \right] = \mathbb{E}_{s \sim \mu} [V^{\pi_{\theta}}(s)]$$

Policy Parametrization

- ▶ Direct parametrization: $\pi_{\theta}(a|s) = \theta_{s,a}$, where $\theta_{s,a} \geq 0$ and $\sum_{a \in \mathcal{A}} \theta_{s,a} = 1$.
- ▶ Softmax policy: $\pi_{\theta}(a|s) = \frac{\exp(\theta_{s,a})}{\sum_{a' \in \mathcal{A}} \exp(\theta_{s,a'})}$.
- ▶ Log-linear policy: $\pi_{\theta}(a|s) = \frac{\exp(\boldsymbol{\theta} \cdot \boldsymbol{\phi}(s,a))}{\sum_{a' \in \mathcal{A}} \exp(\boldsymbol{\theta} \cdot \boldsymbol{\phi}(s,a'))}$.
- ▶ Neural softmax policy: $\pi_{\theta}(a|s) = \frac{\exp(f_{\boldsymbol{\theta}}^{NN}(s,a))}{\sum_{a' \in \mathcal{A}} \exp(f_{\boldsymbol{\theta}}^{NN}(s,a'))}$.

Policy Gradient Theorem

Policy Gradient Theorem

$$\nabla_{\theta} J(\pi_{\theta}) = \mathbb{E}_{\tau \sim p_{\theta}} \left[\sum_{t=0}^{\infty} \gamma^t [Q^{\pi_{\theta}}(s_t, a_t) - b(s_t)] \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \right]$$

$$\nabla_{\theta} J(\pi_{\theta}) = \frac{1}{1-\gamma} \mathbb{E}_{s \sim d_{\mu}^{\pi_{\theta}}, a \sim \pi_{\theta}(\cdot | s)} [Q^{\pi_{\theta}}(s, a) \nabla_{\theta} \log \pi_{\theta}(a | s)]$$

- ▶ Unbiased policy gradient estimators can suffer from high variance
- ▶ Reduce the variance of stochastic policy gradient by using a baseline
- ▶ Can apply other variance reduction techniques

Exercise: Policy Gradient under Tabular Parametrization

Compute policy gradient under the direct and softmax parametrization in the tabular setting.

Direct parametrization

$$\pi_{\theta}(a|s) = \theta_{s,a},$$

where $\theta_{s,a} \geq 0$ and $\sum_{a \in \mathcal{A}} \theta_{s,a} = 1$.

Softmax policy

$$\pi_{\theta}(a|s) = \frac{\exp(\theta_{s,a})}{\sum_{a' \in \mathcal{A}} \exp(\theta_{s,a'})}$$

where $\theta \in \mathbb{R}^{|\mathcal{A}| \times |\mathcal{S}|}$.

$$\frac{\partial J(\pi_{\theta})}{\partial \theta_{s,a}} = ??$$

Policy Gradient in Tabular Setting

Direct Parametrization:

$$\pi_{\theta}(a|s) = \theta_{s,a}$$

$$\frac{\partial J(\pi_{\theta})}{\partial \theta_{s,a}} = \frac{1}{1-\gamma} d_{\mu}^{\pi_{\theta}}(s) Q^{\pi_{\theta}}(s, a)$$

Softmax Parametrization:

$$\pi_{\theta}(a|s) \propto \exp(\theta_{s,a})$$

$$\frac{\partial J(\pi_{\theta})}{\partial \theta_{s,a}} = \frac{1}{1-\gamma} d_{\mu}^{\pi_{\theta}}(s) \pi_{\theta}(a|s) A^{\pi_{\theta}}(s, a)$$

Proof:

- ▶ Recall that $\nabla_{\theta} J(\pi_{\theta}) = \frac{1}{1-\gamma} \sum_s d_{\mu}^{\pi_{\theta}}(s) \sum_a Q^{\pi_{\theta}}(s, a) \nabla_{\theta} \pi_{\theta}(a|s)$.
- ▶ Direct case: $\frac{\partial \pi_{\theta}(a|s)}{\partial \theta_{s',a'}} = 1\{s = s', a = a'\}$.
- ▶ Softmax case: $\frac{\partial \pi_{\theta}(a|s)}{\partial \theta_{s',a'}} = \pi_{\theta}(a|s) 1\{s = s', a = a'\} - \pi_{\theta}(a|s) \pi_{\theta}(a'|s) 1\{s = s'\}$.

Toy Example

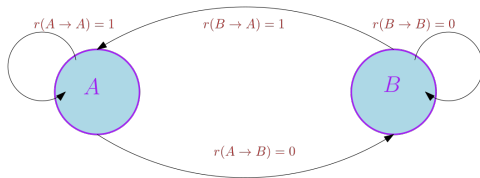


Figure: MDP with 2 states and 2 actions

- ▶ $\mathcal{S} = \{A, B\}$, $\mathcal{A} = \{\text{switch}, \text{stay}\}$,
- ▶ $r(A, \text{switch}) = 0, r(A, \text{stay}) = 1$
 $r(B, \text{switch}) = 1, r(B, \text{stay}) = 0$
- ▶ $P(A|A, \text{switch}) = 0, P(A|A, \text{stay}) = 1$
 $P(B|B, \text{switch}) = 0, P(B|B, \text{stay}) = 1$.
- ▶ Optimal policy:
 $\pi^*(\text{stay}|A) = 1, \pi^*(\text{switch}|B) = 1$.

Question: Does policy gradient method with stepsize small enough converge to π^* by maximizing $V^{\pi_\theta}(A)$ or $V^{\pi_\theta}(B)$?

Toy Example (cont'd)

First consider the direct parametrization with $\theta_1, \theta_2 \in [0, 1]$:

$$\pi(\text{stay}|A) = \theta_1, \quad \pi(\text{switch}|B) = \theta_2$$

- By Bellman equations, we can compute that

$$V^{\pi_\theta}(A) = \frac{\theta_1 - \gamma(\theta_1 - \theta_2)}{(1 - \gamma)[1 - \gamma(\theta_1 - \theta_2)]}, \quad V^{\pi_\theta}(B) = \frac{\theta_2}{(1 - \gamma)[1 - \gamma(\theta_1 - \theta_2)]}.$$

Note both $V^{\pi_\theta}(A)$ and $V^{\pi_\theta}(B)$ are non-concave.

- Maximizing $V^{\pi_\theta}(A)$ with policy gradient method does not always find the optimal policy.
- Maximizing $V^{\pi_\theta}(B)$ with policy gradient method with small constant stepsize can reach the optimal policy within finite steps.
 - Note policy gradient $\frac{\partial V^{\pi_\theta}(B)}{\partial \theta_1} = \frac{\gamma\theta_2}{(1-\gamma)[1-\gamma(\theta_1-\theta_2)]^2}$, $\frac{\partial V^{\pi_\theta}(B)}{\partial \theta_2} = \frac{1-\gamma\theta_1}{(1-\gamma)[1-\gamma(\theta_1-\theta_2)]^2}$.
 - The sequences of $\{\theta_1^t\}, \{\theta_2^t\}$ keep increasing strictly until reaching $(1, 1)$.

Toy Example (cont'd)

Now consider the special parametrization with $\theta \in [0, 1]$:

$$\pi(\text{stay}|A) = \theta, \quad \pi(\text{switch}|B) = \theta$$

- By Bellman equations, we can compute that

$$V^{\pi_\theta}(A) = V^{\pi_\theta}(B) = \frac{\theta}{1 - \gamma}.$$

Note both $V^{\pi_\theta}(A)$ and $V^{\pi_\theta}(B)$ are concave.

- Maximizing $V^{\pi_\theta}(A)$ or $V^{\pi_\theta}(B)$ with policy gradient method with small constant stepsize can reach the optimal policy within finite steps.

Toy Example (cont'd)

We now consider another special parametrization with $\theta \in [0, \frac{5}{4}]$:

$$\pi(\text{stay}|A) = \pi(\text{switch}|B) = \left(\theta - \frac{1}{4}\right)^2.$$

- By Bellman equations, we can compute that

$$V^{\pi_\theta}(A) = V^{\pi_\theta}(B) = \frac{\left(\theta - \frac{1}{4}\right)^2}{1 - \gamma}.$$

Note both $V^{\pi_\theta}(A)$ and $V^{\pi_\theta}(B)$ are convex.

- Maximizing $V^{\pi_\theta}(A)$ or $V^{\pi_\theta}(B)$ with policy gradient method does not always converge to the optimal policy. If initialized at $0 \leq \theta^0 < \frac{1}{4}$, policy gradient method converges to $(0, 0)$.

Toy Example (cont'd)

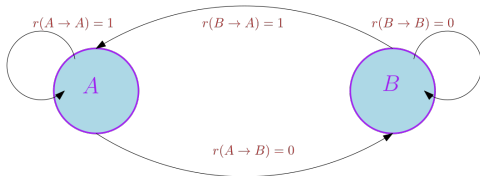


Figure: MDP with 2 states and 2 actions

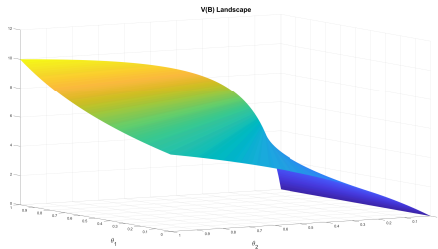


Figure: $V^\pi(B)$ under direct parametrization

Toy Example (cont'd)

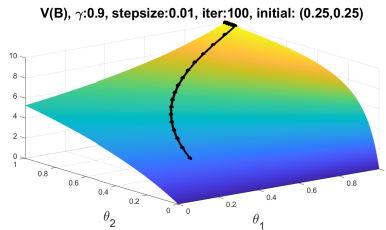
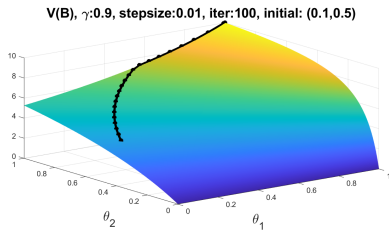
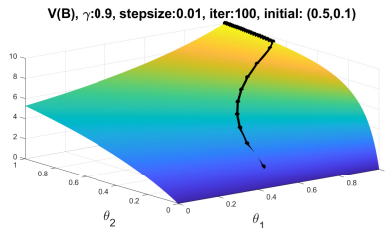
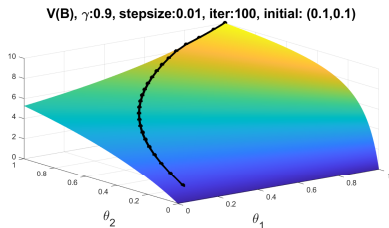


Figure: PG with different initial points

Toy Example (cont'd)

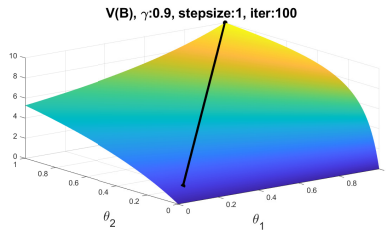
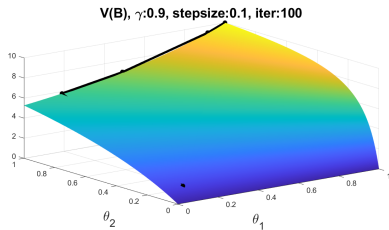
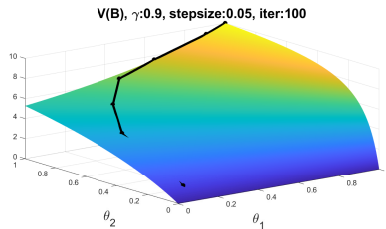
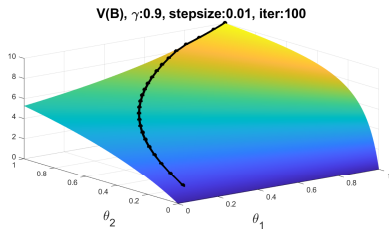


Figure: PG with different stepsizes

Observations so far

- ▶ The optimization landscape varies under different parameterizations.
- ▶ Under restricted policy class even if it contains the optimal policy, policy gradient method may not always find the optimal policy.
- ▶ With direct parametrization, and appropriate initial state distribution (e.g. $\mu(B) > 0$), policy gradient method converges to the optimal solution. Moreover, it converges faster with larger stepsize.

This Lecture: Fundamental Questions

Question 1

Can/When (stochastic) PG methods converge to the globally optimal policy?

- **Optimization wisdom:** GD/SGD could converge to the global optima for “convex-like” functions:

$$J(\pi^*) - J(\pi) = O(\|\nabla J(\pi)\|^p)$$

Question 2

Why/How does NPG perform better than PG?

- **Optimization wisdom:** NPG on parameter space = Mirror Descent on policy space

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Performance Difference Lemma (PDL)

Performance Difference Lemma (Kakade and Langford, 2002)

For any two policy π, π' :

$$J(\pi) - J(\pi') = \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d_{\mu}^{\pi}, a \sim \pi(\cdot|s)} \left[A^{\pi'}(s, a) \right]$$

- ▶ Here $d_{\mu}^{\pi}(s) = (1 - \gamma) \mathbb{E}[\sum_{t=0}^{\infty} \gamma^t \mathbf{1}_{\{s_t=s\}} | s_0 \sim \mu, \pi]$ is the state visitation distribution.
- ▶ Here $A^{\pi}(s, a) = Q^{\pi}(s, a) - V^{\pi}(s)$ is the advantage function.
- ▶ Can be used to show policy improvement theorem for policy iteration (self-exercise).
- ▶ Can also be used to show policy gradient theorem (self-exercise).
- ▶ Proof follows from definition of value functions.

Proof of PDL

Proof.

$$\begin{aligned} V^\pi(s) - V^{\pi'}(s) &= \mathbb{E}_{\tau \sim p_\pi(\tau)} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) | s_0 = s \right] - V^{\pi'}(s) \\ &= \mathbb{E}_{\tau \sim p_\pi(\tau)} \left[\sum_{t=0}^{\infty} \gamma^t (r(s_t, a_t) + V^{\pi'}(s_t) - V^{\pi'}(s_t)) | s_0 = s \right] - V^{\pi'}(s) \\ &= \mathbb{E}_{\tau \sim p_\pi(\tau)} \left[\sum_{t=0}^{\infty} \gamma^t (r(s_t, a_t) + \gamma V^{\pi'}(s_{t+1}) - V^{\pi'}(s_t)) | s_0 = s \right] \\ &= \mathbb{E}_{\tau \sim p_\pi(\tau)} \left[\sum_{t=0}^{\infty} \gamma^t (r(s_t, a_t) + \gamma \mathbb{E}_{s_{t+1} \sim P(\cdot | s_t, a_t)} [V^{\pi'}(s_{t+1})] - V^{\pi'}(s_t)) | s_0 = s \right] \\ &= \mathbb{E}_{\tau \sim p_\pi(\tau)} \left[\sum_{t=0}^{\infty} \gamma^t (Q^{\pi'}(s_t, a_t) - V^{\pi'}(s_t)) | s_0 = s \right] \\ &= \mathbb{E}_{\tau \sim p_\pi(\tau)} \left[\sum_{t=0}^{\infty} \gamma^t A^{\pi'}(s_t, a_t) | s_0 = s \right] \end{aligned}$$

□

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Key Insight: policy optimization is convex-like in the full policy space

- Performance difference lemma:

$$J(\pi^*) - J(\pi) = \frac{1}{1 - \gamma} \sum_s d_{\mu}^{\pi^*}(s) \sum_a \pi^*(a|s) A^{\pi}(s, a)$$

- Policy gradient theorem (tabular setting):

$$\frac{\partial J(\pi)}{\partial \pi(a|s)} = \frac{1}{1 - \gamma} d_{\mu}^{\pi}(s) Q^{\pi}(s, a) \quad (\text{direct parametrization})$$

$$\frac{\partial J(\pi)}{\partial \pi(a|s)} = \frac{1}{1 - \gamma} d_{\mu}^{\pi}(s) \pi(a|s) A^{\pi}(s, a) \quad (\text{softmax parametrization})$$

- This seems to imply gradient dominance type properties:

$$J(\pi^*) - J(\pi) = O(\|\nabla J(\pi)\|),$$

which is crucial to ensure global optimality.

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Policy Optimization under Direct Parametrization

We first consider the direct parametrization in the tabular setting.

Policy Optimization under Direct Parametrization

$$\max_{\pi \in \Delta(\mathcal{A})^{|\mathcal{S}|}} J(\pi) := \mathbb{E}_{s \sim \mu}[V^\pi(s)]$$

where $\Delta(\mathcal{A})^{|\mathcal{S}|} = \{\pi : \pi(a|s) \geq 0, \sum_{a \in \mathcal{A}} \pi(a|s) = 1, \forall s\}$. For simplicity, denote as Π .

- If $\pi \in \Pi$ is optimal, then it satisfies the **first-order optimality condition**:

$$\langle \bar{\pi} - \pi, \nabla J(\pi) \rangle \leq 0, \forall \bar{\pi} \in \Pi$$

or equivalently, $\max_{\bar{\pi} \in \Pi} \langle \bar{\pi} - \pi, \nabla J(\pi) \rangle = 0$.

- Q: Does the reverse statement hold?

Gradient Dominance Property

Gradient Mapping Domination

$$J(\pi^\star) - J(\pi) \leq \left\| \frac{d_\mu^{\pi^\star}}{d_\mu^\pi} \right\|_\infty \max_{\bar{\pi} \in \Pi} \langle \bar{\pi} - \pi, \nabla J(\pi) \rangle$$

- ▶ Any first-order stationary point is thus globally optimal.
- ▶ The term $\left\| \frac{d_\mu^{\pi^\star}}{d_\mu^\pi} \right\|_\infty$ is called the **distribution mismatch coefficient**, which captures the hardness of the exploration problem. Note that in the vanishing gradient example from last lecture, this coefficient can be very exponentially large.
- ▶ Note that $\max_\pi \left\| \frac{d_\mu^{\pi^\star}}{d_\mu^\pi} \right\|_\infty \leq \frac{1}{1-\gamma} \left\| \frac{d_\mu^{\pi^\star}}{\mu} \right\|_\infty$ since $\forall \pi, d_\mu^\pi(s) \geq (1-\gamma)\mu(s)$.
- ▶ Proof follows by combining performance difference lemma and policy gradient theorem.

Proof of Gradient Dominance

Proof.

$$\begin{aligned} J(\pi^*) - J(\pi) &= \frac{1}{1-\gamma} \sum_s d_\mu^{\pi^*}(s) \sum_a \pi^*(a|s) A^\pi(s, a) \\ &= \frac{1}{1-\gamma} \sum_s \frac{d_\mu^{\pi^*}(s)}{d_\mu^\pi(s)} d_\mu^\pi(s) \sum_a \pi^*(a|s) A^\pi(s, a) \\ &\leq \frac{1}{1-\gamma} \left\| \frac{d_\mu^{\pi^*}}{d_\mu^\pi} \right\|_\infty \cdot \max_{\bar{\pi} \in \Pi} \sum_{s,a} d_\mu^\pi(s) \bar{\pi}(a|s) A^\pi(s, a) \\ &= \frac{1}{1-\gamma} \left\| \frac{d_\mu^{\pi^*}}{d_\mu^\pi} \right\|_\infty \cdot \max_{\bar{\pi} \in \Pi} \sum_{s,a} d_\mu^\pi(s) (\bar{\pi}(a|s) - \pi(a|s)) A^\pi(s, a) \\ &= \frac{1}{1-\gamma} \left\| \frac{d_\mu^{\pi^*}}{d_\mu^\pi} \right\|_\infty \cdot \max_{\bar{\pi} \in \Pi} \sum_{s,a} d_\mu^\pi(s) (\bar{\pi}(a|s) - \pi(a|s)) Q^\pi(s, a) \\ &= \left\| \frac{d_\mu^{\pi^*}}{d_\mu^\pi} \right\|_\infty \cdot \max_{\bar{\pi} \in \Pi} \langle \bar{\pi} - \pi, \nabla J(\pi) \rangle \end{aligned}$$

□

Projected Policy Gradient Method

Projected Policy Gradient Method

$$\pi_{t+1} = \mathbf{Proj}_{\Pi}(\pi_t + \eta \nabla J(\pi_t))$$

where the projection $\mathbf{Proj}_{\Pi}(\pi) = \arg \min_{\pi' \in \Pi} \|\pi - \pi'\|_2^2$.

- ▶ Take a gradient ascent step and project onto the simplex set (can be computed efficiently)
- ▶ *Generalized gradient mapping*: $G(\pi_t) = \frac{1}{\eta} (\pi_{t+1} - \pi_t)$, or equivalently, $\pi_{t+1} = \pi_t + \eta G(\pi_t)$.
- ▶ If π is optimal, then $G(\pi) = 0$. (why?)
- ▶ Convergence on gradient mapping [4]: If $J(\pi)$ is L -smooth, then we have

$$\min_{t \leq T} \|G(\pi_t)\|_2^2 \leq \frac{2L(J(\pi^*) - J(\pi_0))}{T}.$$

Global Convergence of Projected Policy Gradient Method

Theorem (Agarwal et al., 2020 [1])

Assume access to exact gradient. Let $\eta = \frac{(1-\gamma)^3}{2\gamma|\mathcal{A}|}$. Then

$$J(\pi^*) - \max_{t \leq T} J(\pi_t) \leq \frac{8 \sqrt{\gamma|\mathcal{S}||\mathcal{A}|}}{(1-\gamma)^3 \sqrt{T}} \left\| \frac{d_\mu^{\pi^*}}{\mu} \right\|_\infty$$

Proof sketch:

- Show that the objective $J(\pi)$ is L -smooth with $L = \frac{2\gamma|\mathcal{A}|}{(1-\gamma)^3}$ and $J(\pi) \leq \frac{1}{1-\gamma}$.
- Invoke convergence on gradient mapping: $\min_{t \leq T} \|G(\pi_t)\|_2^2 \leq \frac{2L(J(\pi^*) - J(\pi_0))}{T}$.
- Invoke the relationship between gradient mapping and approximation of stationary point [4]:

$$\max_{\bar{\pi} \in \Pi} \langle \bar{\pi} - \pi_{t+1}, \nabla J(\pi_{t+1}) \rangle \leq (1 + L\eta) \cdot \|G(\pi_t)\|_2 \cdot \|\pi_{t+1} - \pi_t\|_2.$$

- Use the gradient dominance for global convergence.

Global Convergence of Projected Policy Gradient Method (cont'd)

Theorem (Agarwal et al., 2020 [1])

Assume access to exact gradient. Let $\eta = \frac{(1-\gamma)^3}{2\gamma|A|}$. Then

$$J(\pi^*) - \max_{t \leq T} J(\pi_t) \leq \frac{8 \sqrt{\gamma|\mathcal{S}||A|}}{(1-\gamma)^3 \sqrt{T}} \left\| \frac{d_{\mu}^{\pi^*}}{\mu} \right\|_{\infty}$$

Remarks

- ▶ Large constants in the bound
- ▶ Slow rate in T
- ▶ Analysis can be refined with improved convergence rate of $O\left(\frac{1}{T}\right)$ using Nesterov's result in [5, 10]
- ▶ Recall in tabular setting, VI or PI converges linearly, which is much faster.
- ▶ Q: Is it possible to achieve faster rate for projected policy gradient?

A Closer Look at Exact PG Method

- The projected PG update can also be viewed as

$$\begin{aligned}\pi_{t+1} &:= \mathbf{Proj}_{\Pi}(\pi_t + \eta \nabla J(\pi_t)) \\ &= \arg \max_{\pi \in \Pi} \left\{ \langle \nabla J(\pi_t), \pi \rangle - \frac{1}{2\eta} \|\pi - \pi_t\|_2^2 \right\} \\ &= \arg \max_{\pi \in \Pi} \left\{ \langle Q^{\pi_t}, \pi \rangle_{d_{\mu}^{\pi_t}} - \frac{1}{2\eta'} \|\pi - \pi_t\|_2^2 \right\}\end{aligned}$$

where $\frac{\partial J(\pi)}{\partial \pi(a|s)} = \frac{1}{1-\gamma} d_{\mu}^{\pi}(s) Q^{\pi}(s, a)$ and $\langle \cdot, \cdot \rangle_{d_{\mu}^{\pi}}$ is the reweighted inner product by d_{μ}^{π} .

- Policy gradient method can be viewed as an approximation of weighted policy iteration.

From Gradient Descent to Mirror Descent: Exploiting the non-Euclidean Geometry

A natural idea to adapt PG under the simplex is to exploit other divergence to regularize the policy:

Policy Mirror Descent

$$\pi_{t+1} = \arg \max_{\pi \in \Pi} \left\{ \langle Q^{\pi_t}, \pi \rangle_{d_\mu^{\pi_t}} - \frac{1}{\eta_t} \sum_s d_\mu^{\pi_t}(s) D(\pi(\cdot|s), \pi_t(\cdot|s)) \right\}$$

- **Projected Q-Ascent:** set $D(p, q) = \frac{1}{2} \|p - q\|_2^2$

$$\pi_{t+1}(\cdot|s) = \mathbf{Proj}_{\Delta(\mathcal{A})}(\pi_t(\cdot|s) - \eta_t Q^{\pi_t}(s, \cdot)), \forall s \in \mathcal{S}$$

- **Exponentiated Q-Ascent:** set $D(p, q) = KL(p||q) = \sum_a p_a \log \frac{p_a}{q_a}$

$$\pi_{t+1}(a|s) = \pi_t(a|s) \frac{\exp(\eta_t Q^t(s, a))}{\sum_{a'} \pi_t(a'|s) \exp(\eta_t Q^t(s, a'))}, \forall a \in \mathcal{A}, s \in \mathcal{S}$$

- This is akin to **natural policy gradient** under softmax parameterization.
- As $\eta_t \rightarrow \infty$, this also reduces to the policy iteration update.

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Natural Policy Gradient Method

Natural Policy Gradient (Kakade, 2002)

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t + \eta (F_{\boldsymbol{\theta}_t})^\dagger \nabla_{\boldsymbol{\theta}} J(\pi_{\boldsymbol{\theta}_t}),$$

or equivalently,

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t + \frac{\eta}{1 - \gamma} \mathbf{w}^*(\boldsymbol{\theta}_t),$$

where $\mathbf{w}^*(\boldsymbol{\theta})$ is the solution to the least square problem

$$\mathbf{w}^*(\boldsymbol{\theta}) \in \arg \min_{\mathbf{w}} \mathbb{E}_{s \sim d_{\mu}^{\pi_{\boldsymbol{\theta}}}, a \sim \pi_{\boldsymbol{\theta}}(\cdot|s)} \left[\left(\mathbf{w}^\top \nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\theta}}(a|s) - A^{\pi_{\boldsymbol{\theta}}}(s, a) \right)^2 \right].$$

- ▶ $F_{\boldsymbol{\theta}}$ is the Fisher Information Matrix: $F_{\boldsymbol{\theta}} = \mathbb{E}_{s \sim d_{\mu}^{\pi_{\boldsymbol{\theta}}}, a \sim \pi_{\boldsymbol{\theta}}(\cdot|s)} \left[\nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\theta}}(a|s) \nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\theta}}(a|s)^\top \right]$.
- ▶ $C^\dagger$ is the Moore–Penrose inverse of the matrix C .

Tabular NPG under Softmax Parameterization

Consider $\pi_{\theta}(a|s) = \frac{\exp(\theta_{s,a})}{\sum_{a'} \exp(\theta_{s,a'})}$ and denote $\pi_t = \pi_{\theta_t}$.

NPG parameter update

$$\theta_{t+1} = \theta_t + \frac{\eta}{1 - \gamma} A^{\pi_t}$$

NPG policy update = policy mirror descent

$$\pi_{t+1}(a|s) = \pi_t(a|s) \frac{\exp(\eta A^{\pi_t}(s, a)/(1 - \gamma))}{Z_t(s)}$$

Convergence of Tabular NPG with Softmax Parametrization

NPG policy update = policy mirror descent

$$\pi_{t+1}(a|s) = \pi_t(a|s) \frac{\exp(\eta A^{\pi_t}(s, a)/(1 - \gamma))}{Z_t(s)}$$

Convergence of Tabular NPG [1]

For any $\eta \geq (1 - \gamma)^2 \log |\mathcal{A}|$ and $T > 0$,

$$J(\pi^*) - J(\pi_T) \leq \frac{2}{(1 - \gamma)^2 T}$$

- ▶ Dimension-free convergence, no dependence on $|\mathcal{A}|, |\mathcal{S}|$.
- ▶ No dependence on distribution mismatch coefficient.
- ▶ Can be viewed as a special case of the convergence result of policy mirror descent.

Proof of Tabular NPG Convergence

Lemma (Policy Improvement)

$$J(\pi) - J(\pi_t) = \frac{1}{\eta} \mathbb{E}_{s \sim d_\mu^\pi} [KL(\pi(\cdot|s) || \pi_t(\cdot|s)) - KL(\pi(\cdot|s) || \pi_{t+1}(\cdot|s)) + \log Z_t(s)].$$

- Recall from **Performance Difference Lemma**:

$$J(\pi) - J(\pi_t) = \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d_\mu^\pi, a \sim \pi(a|s)} [A^{\pi_t}(s, a)].$$

- From the update rule $\pi_{t+1}(a|s) = \pi_t(a|s) \frac{\exp(\eta A^{\pi_t}(s, a)/(1 - \gamma))}{Z_t(s)}$, we have

$$A^{\pi_t}(s, a) = \frac{1 - \gamma}{\eta} \log \frac{\pi_{t+1}(a|s) Z_t(s)}{\pi_t(a|s)}.$$

- Combing these two equations, we have the above lemma.

Proof of Tabular NPG Convergence

Proof.

- ▶ Setting $\pi = \pi^*$ in the previous lemma and telescoping from $t = 0, \dots, T - 1$

$$\frac{1}{T} \sum_{t=0}^{T-1} J(\pi^*) - J(\pi_t) \leq \frac{1}{\eta T} \mathbb{E}_{s \sim d_{\mu}^{\pi^*}} [\text{KL}(\pi^*(\cdot|s) || \pi_0(\cdot|s))] + \frac{1}{\eta T} \sum_{t=0}^T \mathbb{E}_{s \sim d_{\mu}^{\pi^*}} [\log Z_t(s)].$$

- ▶ Setting $\pi = \pi_{t+1}$ in the previous lemma, we have

$$J(\pi_{t+1}) - J(\pi_t) \geq \frac{1}{\eta} \mathbb{E}_{s \sim d_{\mu}^{\pi_{t+1}}} [\log Z_t(s)] \geq \frac{1 - \gamma}{\eta} \mathbb{E}_{s \sim \mu} [\log Z_t(s)] \geq 0, \forall \mu. \text{ (why?)}$$

- ▶ Combining these two equations and the fact that $J(\pi) \geq \frac{1}{1-\gamma}$ implies that

$$\frac{1}{T} \sum_{t=0}^{T-1} J(\pi^*) - J(\pi_t) \leq \frac{\log |\mathcal{A}|}{\eta T} + \frac{1}{(1 - \gamma)^2 T}.$$

□

Outline

Recap: Policy Gradient Methods

Performance Difference Lemma

Global Convergence of Policy Gradient Methods

Exact PG under Direct Parameterization

Global Convergence of Natural Policy Gradient Methods

Tabular NPG with Softmax Parametrization

NPG with Function Approximation

Beyond PG/NPG

NPG with Linear Function Approximation

Consider $\pi_{\theta}(a|s) = \frac{\exp(\theta^\top \phi(s,a))}{\sum_{a'} \exp(\theta^\top \phi(s,a'))}$ and denote $\pi_t = \pi_{\theta_t}$.

NPG parameter update

$$\theta_{t+1} = \theta_t + \frac{\eta}{1-\gamma} \mathbf{w}_t, \text{ where } \mathbf{w}_t = \arg \min_{\mathbf{w}} \mathbb{E}_{s \sim d_{\mu}^{\pi_{\theta}}, a \sim \pi_{\theta}(\cdot|s)} \left[\left(\mathbf{w}^\top \bar{\phi}(s,a) - A^{\pi_{\theta}}(s,a) \right)^2 \right]$$

NPG policy update = approximate policy mirror descent

$$\pi_{t+1}(a|s) = \pi_t(a|s) \frac{\exp(\eta \mathbf{w}_t^\top \phi(s,a)/(1-\gamma))}{Z_t(s)}$$

Convergence of Sample-based NPG with Function Approximation

Sample-based NPG (informal)

- ▶ Use N -step SGD to estimate $\mathbf{w}_t \approx \mathbf{w}^*(\theta_t)$
- ▶ Update $\theta_{t+1} = \theta_t + \frac{\eta}{1-\gamma} \mathbf{w}_t$

Convergence of Sample-based NPG (informal) [1]

$$\mathbb{E} \left[\min_{t < T} J(\pi_{\theta_*}) - J(\pi_{\theta_t}) \right] \leq O \left(\frac{1}{1-\gamma} \sqrt{\frac{2 \log |A|}{T}} + \sqrt{\epsilon_{\text{stat}}} + \sqrt{\epsilon_{\text{bias}}} \right)$$

- ▶ ϵ_{stat} : how close $\mathbf{w}_t$ to a $\mathbf{w}^*(\theta_t)$ (statistical error)
- ▶ ϵ_{bias} : how good the best policy in the class (function approximation error)

NB: Improved linear convergence rate with increasing stepsize [11] or with entropy regularization [2].

Outline

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Popular Benchmarks

Trust Region Policy Optimization

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TRPO (ICML, 2015)

Proximal Policy Optimization Algorithms

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OpenAI

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PPO (arXiv, 2017)

OpenAI implementation: <https://github.com/openai/baselines>

Trust Region Policy Optimization (TRPO)

TRPO (key idea) [7]

$$\begin{aligned} \max_{\boldsymbol{\theta}} \quad & \mathbb{E}_{s \sim d_{\mu}^{\pi_{\boldsymbol{\theta}_t}}, a \sim \pi_{\boldsymbol{\theta}_t}(\cdot | s)} \left[\frac{\pi_{\boldsymbol{\theta}}(a | s)}{\pi_{\boldsymbol{\theta}_t}(a | s)} A^{\pi_{\boldsymbol{\theta}_t}}(s, a) \right], \\ \text{s.t.} \quad & \mathbb{E}_{s \sim d_{\mu}^{\pi_{\boldsymbol{\theta}_t}}} [\text{KL}(\pi_{\boldsymbol{\theta}}(\cdot | s) \| \pi_{\boldsymbol{\theta}_t}(\cdot | s))] \leq \delta. \end{aligned}$$

- ▶ The surrogate objective can be viewed as linear approximation (in π instead of $\boldsymbol{\theta}$) of $J(\pi_{\boldsymbol{\theta}})$:

$$J(\pi) = J(\pi_t) + \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d_{\mu}^{\pi}, a \sim \pi(a | s)} [A^{\pi_t}(s, a)]. \quad (\text{PDL})$$

- ▶ Can be approximated by natural policy gradient step.
- ▶ Use line-search to ensure performance improvement and no constraint violation.
- ▶ Extension to trust regions based on other divergences, e.g., Wasserstein distance [3], Sinkhorn divergence [9], etc.

Proximal Policy Optimization (PPO2)

PPO (key idea) [8]

$$\max_{\theta} \mathbb{E}_{s' \sim d_{\mu}^{\pi_{\theta_t}}, a \sim \pi_{\theta_t}(\cdot|s)} \min \left\{ \frac{\pi_{\theta}(a|s)}{\pi_{\theta_t}(a|s)} A^{\pi_{\theta_t}}(s, a), \text{clip} \left(\frac{\pi_{\theta}(a|s)}{\pi_{\theta_t}(a|s)}; 1 - \epsilon; 1 + \epsilon \right) A^{\pi_{\theta_t}}(s, a) \right\}$$

- PPO penalizes large deviation from the current policy directly inside the objective function through clipping the ratio $\frac{\pi_{\theta}}{\pi_{\theta_t}}$.

$$\text{clip}(x; 1 - \epsilon; 1 + \epsilon) = \begin{cases} 1 - \epsilon, & \text{if } x < 1 - \epsilon \\ 1 + \epsilon, & \text{if } x > 1 + \epsilon \\ x, & \text{otherwise} \end{cases}$$

- Run SGD. No need to deal with the KL divergence and trust region constraint.

Applications using PPO



Robots



Locomotion

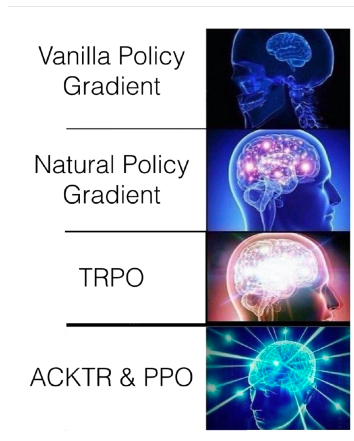


Muti-agent Games

Figure: PPO performs well in many locomotion task and games.

- ▶ https://www.youtube.com/watch?v=hx_bgoTF7bs
- ▶ <https://openai.com/blog/openai-baselines-ppo/>
- ▶ Building block of ChatGPT [6]

From Theory to Practice



Theory



Practice

Figure from Schulman's slide on PPO in 2017.

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